

GLA UNIVERSITY

Application Build and Release Automation Lab

ASSIGNMENT – 3

Linux and Git Version Control

Student: Soyab Khan

Roll Number: 2415001900

Semester: 5th Semester

Project: Online Library Management System

Task 1 – Project Workspace (6 Marks)

Question: Create the project workspace with src, docs, config, backup and library_info.txt. Demonstrate directory navigation, listing, file contents and directory structure.

```

C:\ Command Prompt
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin>mkdir Soyab-GLA-LibrarySystem

C:\Users\admin>cd Soyab-GLA-LibrarySystem

C:\Users\admin\Soyab-GLA-LibrarySystem>mkdir src docs config backup

C:\Users\admin\Soyab-GLA-LibrarySystem>mkdir Soyab-GLA-LibrarySystem

C:\Users\admin\Soyab-GLA-LibrarySystem>cd Soyab-GLA-LibrarySystem

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>mkdir src docs config backup

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>dir
Volume in drive C has no label.
Volume Serial Number is 22C2-B06E

Directory of C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem

11-09-2026  14:41    <DIR>          .
11-09-2026  14:41    <DIR>          ..
11-09-2026  14:41    <DIR>          backup
11-09-2026  14:41    <DIR>          config
11-09-2026  14:41    <DIR>          docs
11-09-2026  14:41    <DIR>          src
               0 File(s)              0 bytes
               6 Dir(s)  11,590,885,376 bytes free

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>dir
Volume in drive C has no label.
Volume Serial Number is 22C2-B06E

Directory of C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem

11-09-2026  14:42    <DIR>          .
11-09-2026  14:41    <DIR>          ..
11-09-2026  14:41    <DIR>          backup
11-09-2026  14:41    <DIR>          config
11-09-2026  14:41    <DIR>          docs
11-09-2026  14:42          174 library_info.txt
11-09-2026  14:41    <DIR>          src
               1 File(s)             174 bytes
               6 Dir(s)  11,591,053,312 bytes free

```

Screenshot 1 – Project folder creation and directory listing.

```
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>type library_info.txt
Name: Soyab Khan
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th Semester
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>cd
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>cd src
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>cd
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>cd
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>tree
Folder PATH listing
Volume serial number is 22C2-B06E
C:.
No subfolders exist
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>cd ..
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>cd
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>tree
Folder PATH listing
Volume serial number is 22C2-B06E
C:.
|___backup
|___config
|___docs
|___src
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git init
Initialized empty Git repository in C:/Users/admin/Soyab-GLA-LibrarySystem/Soyab-GLA-LibrarySystem/.git/
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git config user.name "Soyab Khan"
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git config user.email "YOUR_EMAIL@gmail.com"
```

Screenshot 2 – library_info.txt, navigation and directory structure.

Task 2 – Git Repository (6 Marks)

Question: Initialize Git, configure username/email, check status, stage files, create the initial commit, modify library_info.txt, add objective and description, stage and create the second commit, and display history.

```
Command Prompt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git init
Reinitialized existing Git repository in C:/Users/admin/Soyab-GLA-LibrarySystem/Soyab-GLA-LibrarySystem/.git/

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git config user.name "Soyab Khan"

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git config user.email "soyab.gla_cs24@gla.ac.in"

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch -M main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add .

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Initial commit"
[main (root-commit) 66606e7] Initial commit
1 file changed, 5 insertions(+)
create mode 100644 library_info.txt
```

Screenshot 3 – Git initialization, configuration, status and staging.

```

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Initial commit"
[main (root-commit) 66606e7] Initial commit
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git log --oneline
66606e7 (HEAD -> main) Initial commit

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch -M main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add .

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
nothing to commit, working tree clean

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git log --oneline
66606e7 (HEAD -> main) Initial commit

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git diff
diff --git a/library_info.txt b/library_info.txt
index dd9400c..71dbc71 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -2,4 +2,9 @@ Name: Soyab Khan
 Roll Number: 2415001900
 Subject: Application Build and Release Automation Lab
 Project Name: Online Library Management System
-Semester: 5th Semester
\ No newline at end of file
+Semester: 5th Semester
+Project Objective:
+To develop an Online Library Management System for managing books and library records efficiently.

```

Screenshot 4 – Initial commit and commit history.

```

+To develop an Online Library Management System for managing books and library records efficiently.
+
+Project Description:
+The system will help manage library books and provide basic book management features.
\ No newline at end of file

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Add project objective and description"
[main 08897d0] Add project objective and description
 1 file changed, 6 insertions(+), 1 deletion(-)

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git log --oneline
08897d0 (HEAD -> main) Add project objective and description
66606e7 Initial commit

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch book-feature

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout book-feature
Switched to branch 'book-feature'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>cd src

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>notepad book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>cd ..

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>type src\book.txt
Book Management Module

Description:
The Book Management Module is used to manage books in the Online Library Management System.

Features:
1. Add new books.
2. Search books by title or author.
3. Update book information.
4. Delete books.

```

Screenshot 5 – Objective/description changes, second commit and book-feature branch creation.

Task 3 – Feature Branch (6 Marks)

Question: Create book-feature, add src/book.txt describing the Book Management Module with at least three features, stage and commit, and demonstrate branch information.

```
+To develop an Online Library Management System for managing books and library records efficiently.
+
+Project Description:
+The system will help manage library books and provide basic book management features.
\ No newline at end of file

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Add project objective and description"
[main 08897d0] Add project objective and description
 1 file changed, 6 insertions(+), 1 deletion(-)

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git log --oneline
08897d0 (HEAD -> main) Add project objective and description
66606e7 Initial commit

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch book-feature

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout book-feature
Switched to branch 'book-feature'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>cd src

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>notepad book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem\src>cd ..

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>type src\book.txt
Book Management Module

Description:
The Book Management Module is used to manage books in the Online Library Management System.

Features:
1. Add new books.
2. Search books by title or author.
3. Update book information.
4. Delete books.
```

Screenshot 6A – book-feature branch creation.

```

5. View available books.
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add src\book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add src\book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Add Book Management Module"
[book-feature e620194] Add Book Management Module
 1 file changed, 11 insertions(+)
 create mode 100644 src/book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch -a
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout main
Switched to branch 'main'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
  book-feature
* main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git merge book-feature
Updating 08897d0..e620194
Fast-forward
 src/book.txt | 11 ++++++++
 1 file changed, 11 insertions(+)
 create mode 100644 src/book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout -b library-update
Switched to a new branch 'library-update'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
  book-feature
* library-update
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Update library objective"
[library-update 377d5ef] Update library objective
 1 file changed, 1 insertion(+), 1 deletion(-)

```

Screenshot 6B – book.txt content, commit, branches, main checkout, merge and library-update branch.

Task 4 – Merge Conflict (8 Marks)

Question: Merge book-feature into main, create library-update, modify the same line differently in the two branches, create and identify the conflict, resolve it manually, then stage, commit and display the final status and history.

```
5. View available books.
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add src\book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add src\book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Add Book Management Module"
[book-feature e620194] Add Book Management Module
 1 file changed, 11 insertions(+)
 create mode 100644 src/book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch -a
* book-feature
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout main
Switched to branch 'main'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
  book-feature
* main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git merge book-feature
Updating 08897d0..e620194
Fast-forward
 src/book.txt | 11 ++++++++
 1 file changed, 11 insertions(+)
 create mode 100644 src/book.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git checkout -b library-update
Switched to a new branch 'library-update'

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git branch
  book-feature
* library-update
  main

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Update library objective"
[library-update 377d5ef] Update library objective
 1 file changed, 1 insertion(+), 1 deletion(-)

Screenshot 7A – Merge and library-update branch.
```

Command Prompt

```
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>notepad library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git add library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   library_info.txt

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Resolve merge conflict"
[main ec7fe3c] Resolve merge conflict

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>type library_info.txt
Name: Soyab Khan
Roll Number: 2415001900
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th Semester
Project Objective:
To provide an efficient system for managing library books and user records.
To efficiently manage library books and records for students and staff.

Project Description:
The system will help manage library books and provide basic book management features.
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
nothing to commit, working tree clean
```

Screenshot 7B – Merge conflict, staging and resolution commit.

```

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git commit -m "Resolve merge conflict"
[main ec7fe3c] Resolve merge conflict

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>type library_info.txt
Name: Soyab Khan
Roll Number: 2415001900
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th Semester
Project Objective:
To provide an efficient system for managing library books and user records.
To efficiently manage library books and records for students and staff.

Project Description:
The system will help manage library books and provide basic book management features.
C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git status
On branch main
nothing to commit, working tree clean

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>git log --oneline --graph --all
*   ec7fe3c (HEAD -> main) Resolve merge conflict
| \
| * 377d5ef (library-update) Update library objective
* | 8926921 Update library objective in main
| /
* e620194 (book-feature) Add Book Management Module
* 08897d0 Add project objective and description
* 66606e7 Initial commit

C:\Users\admin\Soyab-GLA-LibrarySystem\Soyab-GLA-LibrarySystem>

```

Screenshot 8 – Final file, clean status and complete Git history.

Task 5 – Submission (4 Marks)

Question: Submit screenshots of important commands, project directory, Git history, branch information, merge demonstration, and conflict/resolution demonstration.

Submission Checklist

- ✓ Project directory and commands
- ✓ library_info.txt
- ✓ Initial and second commits
- ✓ book-feature branch and book.txt
- ✓ library-update branch
- ✓ Merge conflict and manual resolution
- ✓ Final Git status and history

End of Assignment 3 Practical Record